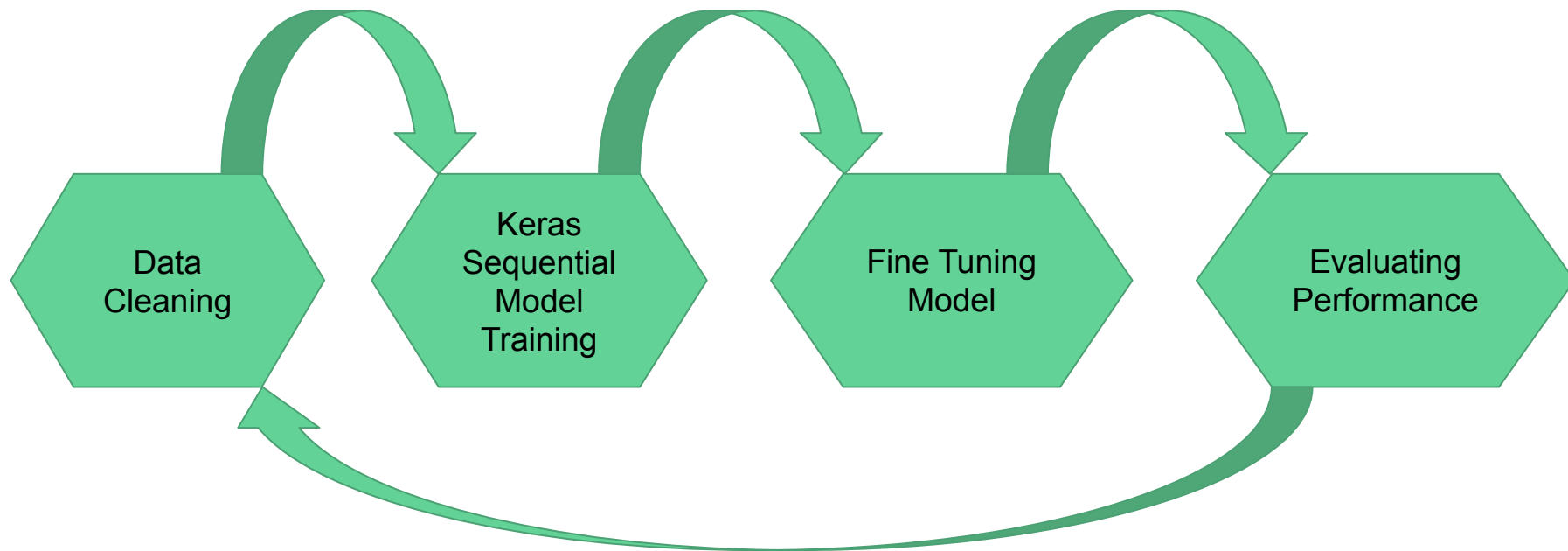
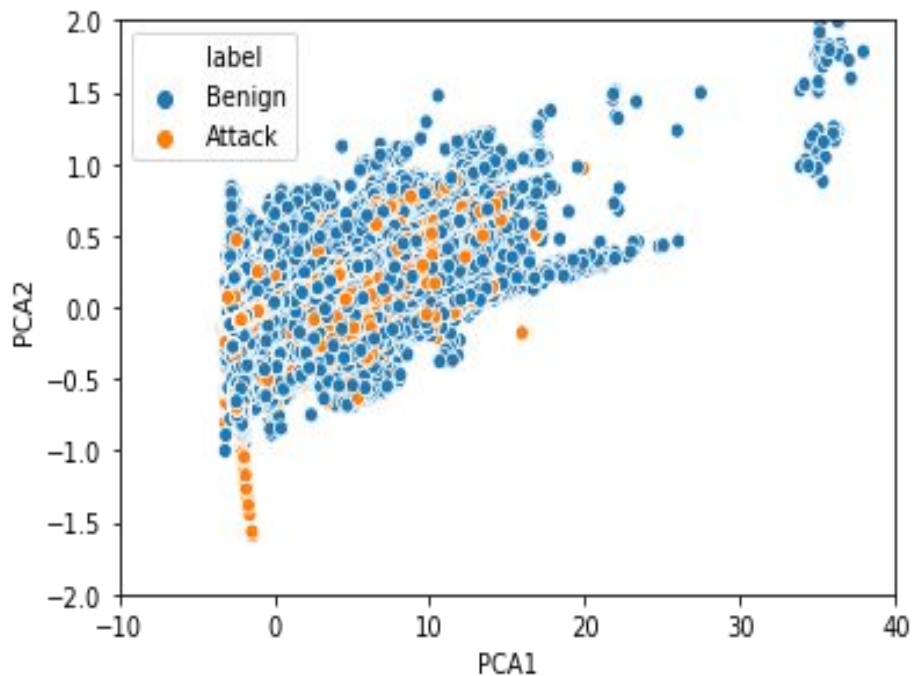


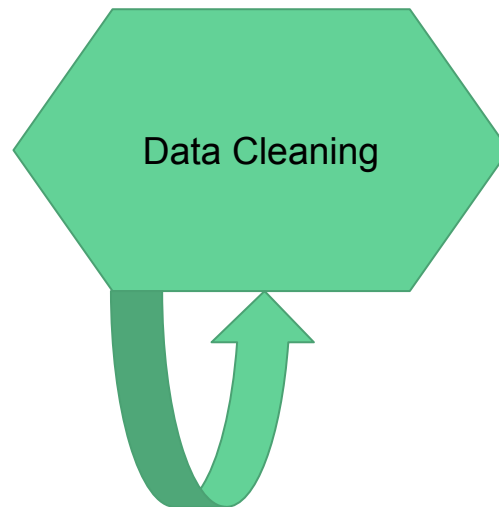
Overview



Data Observations



- Ensure no missing vals
- Explore data balance
- Drop weakly correlated columns
- Scale down data by splitting



Implementing Keras Neural Nets

- Ease of use but also its flexibility
- Support for many model types and settings.
- Can create custom and complex models
- Custom Metrics

Sequential

VS.

Recurrent

- Look at one data point at a time
- Easy, simple to implement

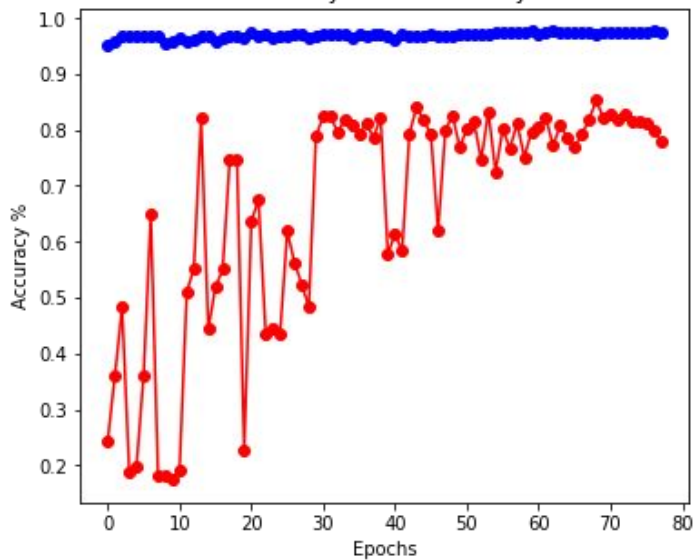
- Look at several data points across a time period
- Process data from initial input to final output
- Loops Info back into the network

Variations Tested

Sequential Model
Binary Classes



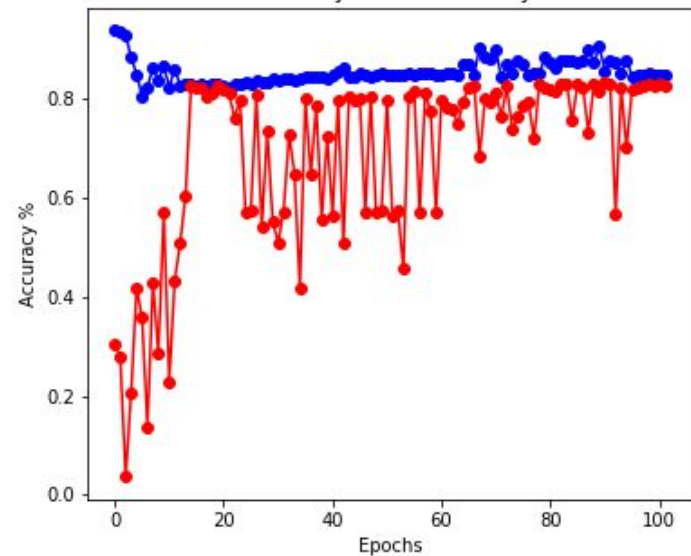
Accuracy Vs. Val Accuracy



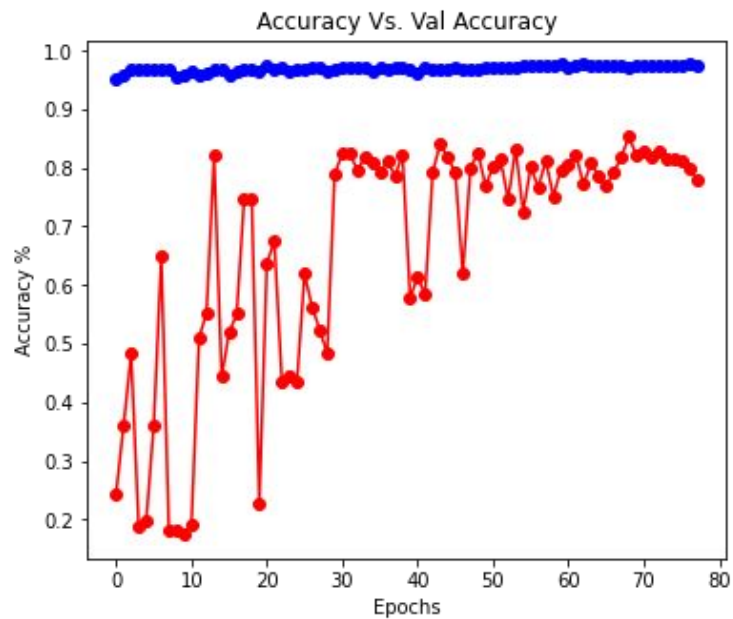
Sequential Model
Multiple Classes



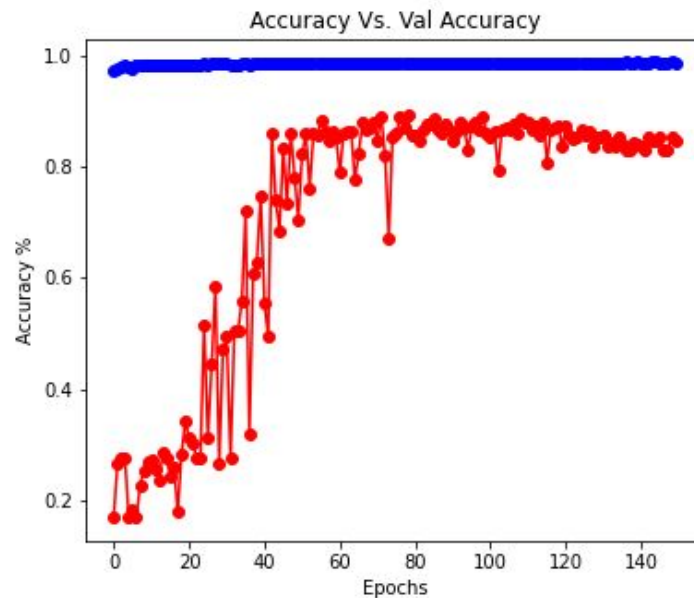
Accuracy Vs. Val Accuracy



Sequential Model Binary Classes



Recurrent Model Binary Classes



Interpreting Results

- Trade off between having to manually allow requests (false positive) vs letting malicious requests through (false negative)
- Choice of model varies with application and available dataset
- More powerful hardware resources will improve training and yield better results

Confusion Matrix

Actual	Predicted		
		Benign	Malware
	Benign	79.70%	3.38%
	Malware	7.51%	9.40%